

MILANO LINATE, Italy

WMO: 160800

Lat: 45.445N

Lon: 9.277E

Elev: 108

StdP: 100.04

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	-2.7	-11.1	1.5	4.2	-8.6	1.8	3.2	9.0	9.0	7.5	8.9	0.9	190	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	33.7	23.8	32.2	22.9	31.0	22.3	25.2	31.2	24.2	30.0	23.3	29.0	2.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.4	28.5	22.3	17.2	27.4	21.4	16.3	26.6	77.5	31.2	73.3	29.9	69.9	28.9	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.4	4.4	DB	-6.4	35.5	2.0	1.3	-7.8	36.5	-9.0	37.2	-10.1	37.9	-11.5	38.8	(4)
(5)			WB	-7.0	26.1	2.0	1.4	-8.5	27.1	-9.6	28.0	-10.8	28.7	-12.2	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	3.7	5.6	10.2	14.1	18.8	23.1	25.2	24.6	20.1	14.9	9.0	4.3	(6)	
	DBStd	8.14	2.64	3.09	3.27	3.04	3.07	3.10	2.31	2.60	2.82	3.15	3.25	2.99	(7)	
	HDD10.0	598	195	125	37	4	0	0	0	0	0	3	56	178	(8)	
	HDD18.3	2068	453	356	252	131	32	3	0	0	14	112	281	436	(9)	
	CDD10.0	2249	1	2	44	126	273	393	472	453	304	156	25	1	(10)	
	CDD18.3	677	0	0	0	3	46	146	213	194	68	7	0	0	(11)	
	CDH23.3	6120	0	0	4	31	359	1336	2133	1801	433	22	0	0	(12)	
	CDH26.7	2248	0	0	0	3	84	489	859	713	97	2	0	0	(13)	
(14) Wind	WSAvg	1.6	1.4	1.5	1.9	2.0	2.0	1.9	1.8	1.5	1.4	1.2	1.3	1.2	(14)	
(15) Precipitation	PrecAvg	971	75	66	82	79	93	71	60	84	72	105	98	60	(15)	
	PrecMax	1413	275	151	202	244	289	153	203	178	214	417	265	156	(16)	
	PrecMin	689	1	1	5	6	4	22	14	1	3	3	1	0	(17)	
	PrecStd	225	65	45	54	55	70	34	49	56	58	93	64	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	18.0	23.2	26.2	31.4	34.7	34.8	35.8	31.0	25.8	18.8	13.8	(19)	
		MCWB	8.2	10.9	13.2	16.4	20.9	24.6	23.5	25.2	21.3	18.3	15.0	8.8	(20)	
	2%	DB	11.2	14.9	20.3	24.0	29.0	32.9	33.2	33.7	29.1	23.3	16.2	11.2	(21)	
		MCWB	7.1	9.2	12.5	15.5	19.4	23.3	23.6	23.8	20.6	17.8	12.9	8.8	(22)	
	5%	DB	9.2	12.8	18.6	22.2	27.0	31.1	32.1	32.0	27.8	21.9	14.9	10.0	(23)	
		MCWB	6.4	7.9	11.9	14.4	18.5	22.3	23.2	22.9	20.0	16.8	12.4	8.1	(24)	
	10%	DB	8.0	10.9	16.7	20.4	25.1	29.7	31.0	30.5	25.9	20.1	13.7	8.9	(25)	
		MCWB	5.9	7.1	11.0	13.7	17.6	21.4	22.4	22.2	18.8	16.2	11.9	7.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	11.5	15.1	17.6	21.7	26.2	26.4	27.0	23.2	19.8	15.6	11.2	(27)	
		MCDB	12.4	16.6	20.1	23.9	29.7	33.3	32.3	33.9	28.6	22.6	17.0	12.3	(28)	
	2%	WB	8.2	9.9	13.6	16.1	20.3	24.5	25.1	25.2	22.1	19.0	14.1	9.7	(29)	
		MCDB	9.6	13.3	18.1	22.3	27.0	30.5	31.0	31.1	26.8	21.8	15.3	10.5	(30)	
	5%	WB	7.1	8.7	12.5	15.2	19.3	23.3	24.1	24.1	21.0	18.1	13.1	8.5	(31)	
		MCDB	8.4	11.3	17.0	20.8	25.4	29.1	30.0	29.6	25.4	20.9	14.3	9.4	(32)	
	10%	WB	6.1	7.7	11.5	14.3	18.3	22.3	23.3	23.1	20.0	17.0	12.1	7.4	(33)	
		MCDB	7.5	9.7	15.7	19.1	23.9	27.8	29.2	28.5	24.1	19.3	13.4	8.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	8.5	10.2	9.8	10.0	10.2	10.5	10.3	9.8	7.9	6.4	6.2	(35)	
		MCDBR	9.8	12.9	14.0	13.1	13.0	11.9	11.4	11.9	11.8	10.6	8.5	8.2	(36)	
	5% WB	MCWBR	6.5	8.0	7.3	6.0	5.6	5.4	4.9	4.8	4.8	5.7	5.5	5.8	(37)	
		MCWBR	7.8	10.8	11.8	11.4	11.5	11.1	10.5	11.1	10.2	8.8	6.4	5.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.349	0.377	0.423	0.465	0.462	0.484	0.465	0.456	0.450	0.433	0.378	0.343	(40)	
		taud	2.379	2.270	2.147	2.062	2.124	2.108	2.168	2.197	2.184	2.187	2.309	2.402	(41)	
	Ebn at Noon	Ebn at Noon	743	785	795	795	812	794	805	795	761	707	692	708	(42)	
		Edh at Noon	77	102	132	156	151	155	144	135	126	110	82	68	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.45	2.24	3.57	4.51	5.58	6.43	6.56	5.55	4.15	2.43	1.44	1.21	(44)	
	RadStd	RadStd	0.21	0.33	0.46	0.50	0.45	0.37	0.40	0.34	0.28	0.38	0.25	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-1.14	-1.61	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		+0.35	N/A	N/A	+0.66	+0.38	+0.29	N/A	N/A	+61	+10

Nomenclature: See separate page